

NUWANANDUN KALHARA

Software Engineering Undergraduate

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PROFESSIONAL SUMMARY

Final-year Software Engineering undergraduate with hands-on experience building full-stack and AI-driven applications using React, Node.js, and Python. Passionate about designing scalable systems and solving real-world problems through modern software engineering practices. Strong interest in backend systems, cloud platforms, and intelligent applications.

EXPERIENCE

Full Stack Developer (Contract) | HardTalk (Pvt) Ltd | November 2025 - Present

- Developing MERN and PERN stack-based internal automation tools and client-facing platforms.
- Built a project management system and a content distribution system for real-time publishing and tracking.
- Delivering tailored software solutions for clients across multiple stacks.

PROJECTS

Adaptive Learning System for Children with ADHD | Final Year Research Project - Ongoing

- Developing an AI-powered adaptive learning platform using multimodal machine learning to detect student distraction in real time and trigger gamified interventions to re-engage learners.

Technologies: Python, React.js, Node.js, FastAPI, WebSockets, SQL Database, Git, AWS

CitizenSafe - Disaster Alert & Community Reporting System | Live Project

- Built a full-stack disaster monitoring platform enabling users to report incidents with geolocation and images, visualize nearby disasters on an interactive map, and receive real-time alerts for community-based verification.

Technologies: React.js, Node.js, MongoDB, Google Maps API, AWS, Firebase, Postman, GitHub Actions

TradexAI - AI-Powered Crypto Trading Intelligence Platform | Live Project

- Built a tri-layer trading signal engine combining Binance data, sentiment analysis, and Gemini AI, achieving 65% accuracy in real-time crypto markets.
- Integrated TradingView charts and a Socket.IO chatbot, with CI/CD deployment on Azure.

Technologies: MERN, Docker, GitHub Actions, Azure, Socket.IO, Gemini AI, Binance API

Restaurant Billing & Monitoring System | Client Project

- Developed a full-stack restaurant billing and monitoring system with real-time sales tracking, inventory management, and remote mobile access for analytics and reports.

Technologies: React.js, Node.js, Firebase, MongoDB Atlas

Bakery Inventory Management System | Client Project

- Built an inventory management system for a bakery to track stock levels, manage product inventory, and generate daily analytics reports, with role-based access control for staff operations.

Technologies: React.js, Node.js, MongoDB Atlas

CitizenGate - Centralized Platform for Government Services

- Developed a government services booking platform with secure authentication, document management, and appointment scheduling.

Technologies: NestJS, PostgreSQL (Supabase), Clerk SDK, React, Vite, Shadcn, Tailwind CSS, Docker, TypeScript

Eshuttle - Smart Campus Transportation System

- Developed a smart transportation management system to digitalize university shuttle operations with live tracking, safety notifications, and real-time analytics.

Technologies: Flutter, React.js, Firebase, Google Maps API, Stripe API

Budget Buddy - Personal Finance & Budgeting Desktop App

- Contributed to a desktop application that helps users manage personal finances by tracking income and expenses, setting monthly budgets, and visualizing spending patterns through an interactive dashboard.

Technologies: Java, Java Swing, JDBC, MySQL, IntelliJ IDEA

AI Powered CV Analyzer

- Built an AI-powered system that analyzes resumes against job descriptions and provides personalized feedback, skill gap analysis, and job match scoring.

Technologies: Python, FastAPI, TypeScript, NLP, Gemini

EDUCATION

BSc (Hons) in Software Engineering | General Sir John Kotelawala Defence University

GCE Advanced Level - Physical Science (3 Passes) | D. S. Senanayake College, Colombo 07

TECHNICAL SKILLS

Languages & Frameworks: React.js, Node.js, Python, TypeScript, Spring Boot, NestJS, Flutter

Databases: MySQL, PostgreSQL, MongoDB

Cloud & DevOps: AWS, Azure, Docker, GitHub Actions

Tools: Git, GitHub, Postman, VS Code, IntelliJ IDEA, Figma

Core Skills: Logical & Analytical Thinking, Effective Communication, Team Collaboration, Adaptability, Time Management

PUBLICATIONS

Real-Time Attention Monitoring and Adaptive Interventions for Children with ADHD: A Systematic Review

Presented at SASIGD 2026 (IEEE SSIT technically co-sponsored), Hyderabad, India.

CERTIFICATIONS

- CCNAv7: Introduction to Networks - Cisco Networking Academy
- Introduction to Packet Tracer - Cisco Networking Academy
- Introduction to LLMs - Sololearn
- Prompt Engineering - Sololearn
- Machine Learning for Beginners - Sololearn
- Python Developer - Sololearn
- Vibe Coding - Sololearn

ACHIEVEMENTS

- Finalist, Tech Triathlon 2025 - Rootcode
- IEEEExtreme 18.0 - Ranked 37th in Sri Lanka (Island) - IEEE (MGA) Student Activities Committee

VOLUNTEERING

- Vice Chair (Public Visibility), LetsTalk 25 - IEEE Young Professionals Sri Lanka
- Logistics Coordinator, LetsTalk 24 - IEEE Young Professionals Sri Lanka
- Student Ambassador, IEEEExtreme 18.0 - General Sir John Kotelawala Defence University

REFERENCES

Dr. LDSB Weerasinghe | Architect Software Engineering, Axiata Digital Labs

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